THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following statements about isotopes is wrong? 1

(a) They differ in mass number **(b)** They have the same atomic number
(c) They have different numbers of electrons **(d)** They have the same chemical properties

2. Assertion (A): The mass of an atom is concentrated in its nucleus. Reason (R): Electrons are much lighter than protons and neutrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. In the Rutherford experiment, if the nucleus had a diameter of 1 cm, the diameter of the atom would be about: 1

(a) 1 m **(b)** 10 m **(c)** 1 km **(d)** 100 km

4. The nucleus of an atom of an element contains 11 protons and 12 neutrons. Its ion with a charge of +1 has electrons: 1

(a) 12 **(b)** 11 **(c)** 10 **(d)** 23

5. Assertion (A): Cobalt-60 is used to treat cancer. Reason (R): It emits gamma radiation that destroys cancer cells. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Which of the following is not allowed by the filling rules? 1

(a) 2, 8, 8, 2 **(b)** 2, 8, 1 **(c)** 2, 8, 3 **(d)** 2, 7, 1

7. An element in the third period has 2 valence electrons. It forms ions with the configuration of: 1

(a) neon **(b)** krypton **(c)** argon **(d)** helium

8. Assertion (A): An atom is electrically neutral. Reason (R): It has equal numbers of protons and neutrons. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. **Assertion (A):** Neon (atomic number 10) is chemically inert. 1
Reason (R): Its outermost shell is completely filled with 8 electrons.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Isotopes of an element have the same chemical properties. 1
Reason (R): Isotopes of an element have different mass numbers.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State why the electron's mass is neglected but its charge is not. 2

12. Why do the atoms of the elements in the same vertical group have similar chemical properties? Use Li, Na, K. 2

13. The nucleus of a hydrogen atom has no neutron. Write the symbol and number of nucleons of the hydrogen atom, deuterium and tritium. 2

14. An element X has mass number 35 and has 18 neutrons. Find its atomic number and the number of valence electrons. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a charged particle moves in a bent path in the electric field in an alpha-scattering experiment, and mention the nature of alpha particles. 3

16. Explain why the atomic number is a better basis for identifying elements than the mass number. 3

17. Explain why tritium has a larger mass than protium but the same chemical reactivity. 3

OR

Explain why an atom of argon is chemically inert but an ion like K^+ is stable although both have the same configuration. 3

18. State the electronic configuration of the elements with Z = 15, 16, 17 and 18 and the number of electrons needed to complete the octet in each. 3

OR

Why do alpha particles have a very short range in air while gamma rays penetrate deeply? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a historic experiment a thin gold foil was bombarded with fast-moving alpha particles. Most particles passed straight through, a small number were deflected through small angles, and very few rebounded almost straight back. 5
(i) Name the scientist who performed this experiment.
(ii) What did the passage of most particles show?
(iii) What did the large deflection of a few particles show about the charge on the centre of the atom?
(iv) State two conclusions about the nucleus drawn from the experiment.

20. Explain the successive discovery of the electron, the proton and the neutron, and how each discovery changed the model of the atom. 5

OR

Explain, in terms of atomic structure, why noble gases are called inert, why they have low boiling points and give their uses. 5

END OF QUESTION PAPER Total: 40 Marks